Database Deployment

This guide covers deploying and managing the PostgreSQL database in production.

Managed PostgreSQL (Recommended)

Provider	Free Tier	Notes
Supabase	Yes	PostgreSQL with REST API, easy setup
Neon	Yes	Serverless PostgreSQL, branching
Railway	$5/month	Integrated with Railway backend
AWS RDS	No	Enterprise-grade, managed backups

Self-Hosted PostgreSQL

Installation

```
# Ubuntu/Debian
sudo apt update
sudo apt install postgresql postgresql-contrib

# Start service
sudo systemctl start postgresql
sudo systemctl enable postgresql
```

Create Database

```
sudo -u postgres psql
CREATE DATABASE kuccps_ticketing;
CREATE USER kuccps_user WITH PASSWORD 'secure-password';
GRANT ALL PRIVILEGES ON DATABASE kuccps_ticketing TO kuccps_user;
\q
```

Migrations

Run migrations on the production database:

```
cd backend
npx prisma migrate deploy
npx prisma generate
```

> **Never run `prisma migrate dev` in production.**

Backups

Automated Backups

Configure automated PostgreSQL backups:

```
# Daily backup script
0 2 * * * pg_dump -U kuccps_user kuccps_ticketing | gzip > /backups/kuccps_$(date
+\%Y\%m\%d).sql.gz
```

pg_dump

```
# Full backup
pg_dump -U kuccps_user -d kuccps_ticketing > backup.sql

# Compressed backup
pg_dump -U kuccps_user -d kuccps_ticketing | gzip > backup.sql.gz

# Specific tables
pg_dump -U kuccps_user -d kuccps_ticketing -t Ticket -t User > tables_backup.sql
```

Restore

```
# Restore from backup
psql -U kuccps_user -d kuccps_ticketing < backup.sql

# Restore compressed
gunzip -c backup.sql.gz | psql -U kuccps_user -d kuccps_ticketing
```

Backup Best Practices

1. **Daily automated backups** with at least 7-day retention
2. **Test restore procedures** monthly
3. **Store backups offsite** (e.g., S3, Google Cloud Storage)
4. **Back up file uploads** separately from database
5. **Document restore procedures** for disaster recovery

Connection Pooling

For high-traffic deployments, use PgBouncer:

```
# Install PgBouncer
sudo apt install pgbouncer

# Configure pool_mode = transaction
# Update DATABASE_URL to use PgBouncer port
```

Monitoring

Monitor database health:

- Connection count
- Query performance
- Disk usage
- Replication lag (if using replicas)

Tools:

- `pg_stat_activity`
- `pg_stat_user_tables`
- `pg_size_pretty(pg_database_size('kuccps_ticketing'))`

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Troubleshooting

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